DAY 10

React.js Introduction — Notes, Programs & Practical Project

Learning Objectives

• Understand React.js and its advantages.

• Learn components, JSX, functional components, and props.

• Create a React application using Vite.

• Understand src and public folders.

• Create Header, Home, and Footer components.

• Run the application locally.

1. What is React.js?

React.js is a JavaScript library used to build interactive user interfaces (UIs). React is maintained by Meta. It is used for websites, dashboards, e-commerce interfaces, and Single Page Applications.

```
My Website
■■■ Header
■■■ Navbar
■■■ ProductList
■    ■■■ ProductCard
■    ■■■ ProductCard
■    ■■■ ProductCard
■■■ Footer
```

2. Advantages of React

Advantage	Explanation
Reusability	Use a component in multiple places.
Component-based	Divide the UI into small parts.
Declarative UI	Describe what the UI should look like.
Efficient updates	React updates the UI when needed.
Ecosystem	Libraries are available for routing, forms, and state management.
Developer tools	Inspect components with React Developer Tools.

3. What is a Component?

A component is a reusable piece of UI. A functional component is a JavaScript function that returns JSX.

```
function Welcome() {
    return <h1>Welcome to React!</h1>;
}
export default Welcome;
```

Use it in another component:

```
import Welcome from "./Welcome";

function App() {
    return <div><Welcome /></div>;
}
export default App;
```

4. What is JSX?

JSX (JavaScript XML) is a syntax extension that lets you write HTML-like markup inside JavaScript.

```
const name = "Vaibhav";

function App() {
    return <h1>Hello, {name}!</h1>;
}
export default App;
// Output: Hello, Vaibhav!
```

JSX rules

• Use className instead of class.

• Close all tags, e.g. .

• Use curly braces {} for JavaScript expressions.

• Return one parent element, or use a Fragment: <> ... </>.

5. Functional Components

Header.jsx

```
function Header() {
    return (
        <header>
            <h1>My Website</h1>
            <nav>
                <a href="/">Home</a>
                <a href="/about">About</a>
            </nav>
        </header>
    );
}
export default Header;
```

Footer.jsx

```
function Footer() {
    return <footer><p>Copyright 2026 My Website</p></footer>;
}
export default Footer;
```

Use components in App.jsx

```
import Header from "./Header";
import Footer from "./Footer";

function App() {
    return (
        <>
            <Header />
            <main>
                <h2>Home Page</h2>
                <p>Welcome to my website.</p>
            </main>
            <Footer />
        </>
    );
}
export default App;
```

6. What are Props?

Props pass data from a parent component to a child component. Treat props as read-only.

Student.jsx

```
function Student(props) {
    return (
        <div>
            <h2>Name: {props.name}</h2>
            <p>Course: {props.course}</p>
        </div>
    );
}
export default Student;
```

App.jsx

```
import Student from "./Student";

function App() {
    return (
        <div>
            <Student name="Vaibhav" course="React.js" />
            <Student name="Rahul" course="Java" />
            <Student name="Priya" course="Python" />
        </div>
    );
}
export default App;
```

Props can also be destructured:

```
function Student({ name, course }) {
    return <div><h2>{name}</h2><p>{course}</p></div>;
}
export default Student;
```

7. Introduction to Vite

Vite is a frontend development tool that provides a development server and builds your application for production.

• Fast development server.

• Quick updates while developing.

• Simple project setup.

• Production build support.

• Works well with React.

8. Create a React Application Using Vite

Step 1 — Check Node.js and npm

```
node -v
npm -v
```

Step 2 — Create the project

```
npm create vite@latest day10-react -- --template react
```

Step 3 — Enter the project folder

```
cd day10-react
```

Step 4 — Install dependencies

```
npm install
```

Step 5 — Start the server

```
npm run dev
```

Open the local URL printed in the terminal, commonly http://localhost:5173/. If Node.js is not installed, download it from the official Node.js website.

9. Understand the Project Structure

```
day10-react/
■■■ node_modules/
■■■ public/
■   ■■■ vite.svg
■■■ src/
■   ■■■ assets/
■   ■■■ App.css
■   ■■■ App.jsx
■   ■■■ index.css
■   ■■■ main.jsx
■■■ index.html
■■■ package.json
■■■ vite.config.js
```

File / Folder	Purpose
src/	React application source code.
src/App.jsx	Main App component.
src/main.jsx	Mounts React into the HTML page.
src/index.css	Global styles.
src/App.css	App-specific styles.
src/assets/	Imported images and other assets.
public/	Static files served directly.
index.html	Main HTML document.
package.json	Dependencies and npm scripts.
node_modules/	Installed packages.

Example: src/main.jsx

```
import { StrictMode } from "react";
import { createRoot } from "react-dom/client";
import App from "./App.jsx";
import "./index.css";

createRoot(document.getElementById("root")).render(
    <StrictMode><App /></StrictMode>
);
```

10. Practical Project — Header, Home, Footer

Create this folder inside src:

```
src/
■■■ components/
■   ■■■ Header.jsx
■   ■■■ Footer.jsx
■■■ App.jsx
■■■ App.css
■■■ index.css
■■■ main.jsx
```

File 1 — src/components/Header.jsx

```
function Header() {
    return (
        <header className="header">
            <h1>My React Website</h1>
            <nav>
                <a href="#home">Home</a>
                <a href="#about">About</a>
                <a href="#contact">Contact</a>
            </nav>
        </header>
    );
}
export default Header;
```

File 2 — src/components/Footer.jsx

```
function Footer({ year, company }) {
    return (
        <footer className="footer">
            <p>
                Copyright &copy; {year} {company}.
                All rights reserved.
            </p>
        </footer>
    );
}
export default Footer;
```

File 3 — src/App.jsx

```
import Header from "./components/Header";
import Footer from "./components/Footer";
import "./App.css";

function Home({ name, course }) {
    return (
        <main className="home" id="home">
            <h2>Welcome, {name}!</h2>
            <p>You are learning {course}.</p>

            <section id="about">
                <h3>About This Project</h3>
                <p>This website is built using React components and props.</p>
            </section>

            <section id="contact">
                <h3>Contact</h3>
                <p>Learn React one concept at a time.</p>
            </section>
        </main>
    );
}

function App() {
    return (
        <>
            <Header />
            <Home name="Vaibhav" course="React.js" />
            <Footer year={2026} company="My React Website" />
        </>
    );
}
export default App;
```

File 4 — src/App.css

```
.header {
    background: #17324d;
    color: white;
    padding: 20px;
    text-align: center;
}
.header nav {
    display: flex;
    justify-content: center;
    gap: 20px;
}
.header a { color: white; text-decoration: none; }
.home {
    max-width: 800px;
    min-height: 60vh;
    margin: 30px auto;
    padding: 20px;
}
.home section { margin-top: 30px; }
.footer {
    background: #eeeeee;
    padding: 20px;
    text-align: center;
}
```

File 5 — src/index.css

```
* { box-sizing: border-box; }

body {
    margin: 0;
    font-family: Arial, sans-serif;
    color: #222;
    line-height: 1.6;
}

a { cursor: pointer; }
```

11. Run the Application

• Save all files using Ctrl + S.

• In the VS Code terminal, run: npm run dev.

• Open the local URL printed by Vite in your browser.

• You should see the header, welcome message, About section, Contact section, and footer.

Expected page content

```
My React Website
Home | About | Contact

Welcome, Vaibhav!
You are learning React.js.

About This Project
This website is built using React components and props.

Contact
Learn React one concept at a time.

Copyright © 2026 My React Website. All rights reserved.
```

12. How Data Flows Between Components

```
App
■■■ Header
■■■ Home
■    ■■■ name = "Vaibhav"
■    ■■■ course = "React.js"
■■■ Footer
    ■■■ year = 2026
    ■■■ company = "My React Website"
```

App is the parent component. Header, Home, and Footer are rendered inside it. Props pass data from parent to child.

13. Practice Tasks

Task	Requirement	Concept
1	Display your name in the header.	JSX
2	Create an About.jsx component.	Components
3	Display a student's name and age using props.	Props
4	Create a reusable StudentCard component.	Reusability
5	Display the current year in the footer.	JS expressions
6	Combine Header, Home, and Footer in App.jsx.	Imports/exports
7	Improve the website design with CSS.	CSS

Bonus Challenge — StudentCard.jsx

```
function StudentCard({ name, course, age }) {
    return (
        <div className="student-card">
            <h2>{name}</h2>
            <p>Course: {course}</p>
            <p>Age: {age}</p>
        </div>
    );
}
export default StudentCard;
```

Use StudentCard in App.jsx

```
<StudentCard name="Vaibhav" course="React.js" age={22} />
<StudentCard name="Rahul" course="Java" age={23} />
```

14. Day 10 Completion Checklist

• ■ Understand React.js and its purpose.

• ■ Understand components and JSX.

• ■ Create functional components.

- ■ Pass data using props.
- ■ Create a React project using Vite.
- ■ Understand src and public folders.
- ■ Create Header, Home, and Footer components.
- ■ Run the app locally.

Recommended next topic: useState, event handling, and conditional rendering.